

# Day 13 Student Handout

Geometry Summer Credit | Sine, Cosine, and Tangent Ratios

|       |       |         |
|-------|-------|---------|
| Name: | Date: | Period: |
|-------|-------|---------|

## Learning Targets

I can label opposite, adjacent, and hypotenuse from a reference angle. I can use sine, cosine, and tangent to find missing side lengths. I can use inverse trig to find missing angle measures.

## Lesson 13A: SOH-CAH-TOA

### Warm-Up

|    |                                                                |
|----|----------------------------------------------------------------|
| 1. | A 30-60-90 triangle has short leg 6. Find the hypotenuse.      |
| 2. | A 45-45-90 triangle has leg 8. Find the hypotenuse.            |
| 3. | Find the missing leg if the hypotenuse is 13 and one leg is 5. |

### Concept Launch

#### Trig Ratios

From a chosen acute angle: sine = opposite / hypotenuse, cosine = adjacent / hypotenuse, and tangent = opposite / adjacent. The hypotenuse is always across from the right angle.

| Problem                                                                                                                                           | Work / notes                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| For angle A in a right triangle, the opposite side is 9, adjacent side is 12, and hypotenuse is 15. Write $\sin(A)$ , $\cos(A)$ , and $\tan(A)$ . | $\sin(A) =$<br>$\cos(A) =$<br>$\tan(A) =$ |
| A right triangle has angle 35 degrees and hypotenuse 20. The missing side is opposite the angle. Set up and solve.                                | Ratio:<br>Equation:<br>Answer:            |
| A right triangle has angle 42 degrees and adjacent side 11. Find the opposite side.                                                               | Ratio:<br>Equation:<br>Answer:            |

# Day 13 Student Handout

Lesson 13B | Inverse Trig and Applications

## Lesson 13B: Finding Angles

### Inverse Trig

Use inverse sine, inverse cosine, or inverse tangent when you know two side lengths and need an angle. Choose the inverse ratio that matches the two sides you know.

| Problem                                                                                              | Work / notes                                                   |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| A right triangle has opposite side 7 and hypotenuse 18 from angle A. Find angle A.                   | Ratio:<br>$\sin(A) = 7/18$<br>$A = \sin^{-1}(7/18)$<br>Answer: |
| A right triangle has adjacent side 10 and opposite side 6 from angle B. Find angle B.                | Ratio:<br>$\tan(B) = 6/10$<br>$B = \tan^{-1}(6/10)$<br>Answer: |
| A ladder is 15 ft long and reaches 12 ft up a wall. Find the angle the ladder makes with the ground. | Known sides from ground angle:<br>Ratio:<br>Answer:            |

### Practice

|    |                                                                                    |
|----|------------------------------------------------------------------------------------|
| 4. | Label the ratio: opposite over hypotenuse is ____.                                 |
| 5. | A right triangle has angle 28 degrees and hypotenuse 16. Find the opposite side.   |
| 6. | A right triangle has angle 50 degrees and adjacent side 9. Find the opposite side. |
| 7. | A right triangle has opposite side 5 and adjacent side 12. Find the angle.         |